

Sistem Terdistribusi

IF2228

02: Networking

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Sistem Terdistribusi

IF2228

Pengampu

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Semester Gasal 2026/2027



Today...

- Announcements:

- ...
-

- Last Session:

- Introduction

- Today's Session :

- Network Types (*Sections 3.1 and 3.2 in Coulouris et al.*)
- Networking Principles: Layering and Encapsulation (*Section 3.3 in Coulouris et al.*)





So, What is a Distributed System?

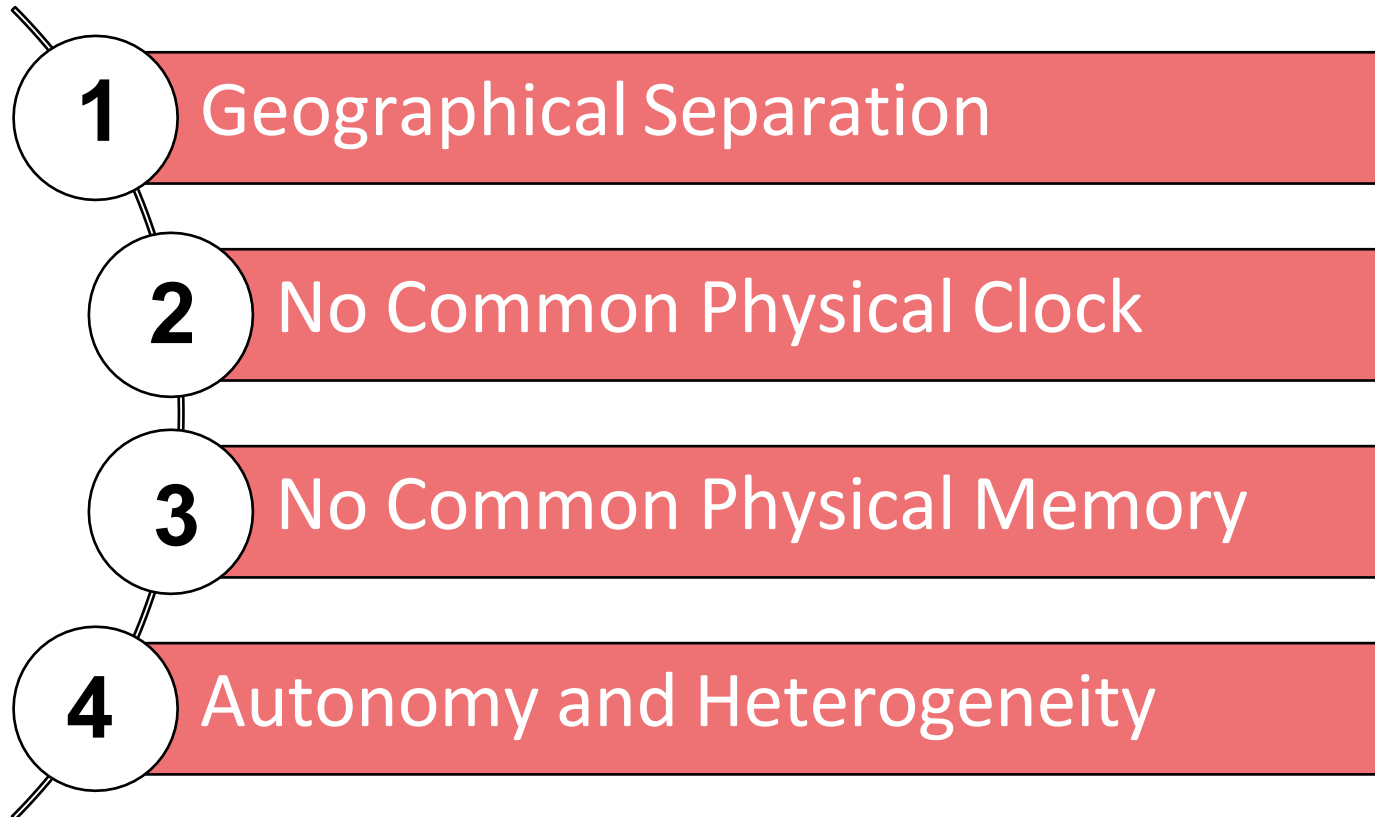
A distributed system is:

*A collection of **independent** computers that appear to its users as **a single coherent system***

*One in which components located at **networked computers** communicate and coordinate their actions only by **passing messages***

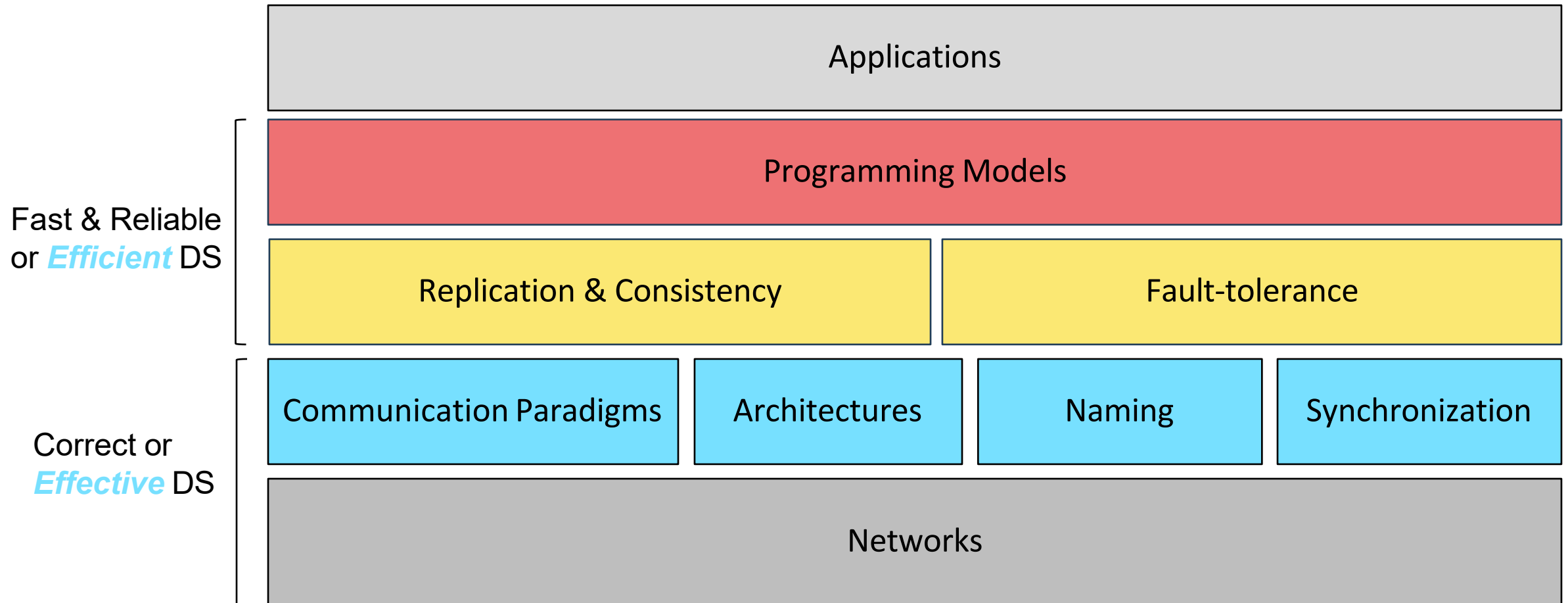
Features

- Distributed Systems imply *four* main features:

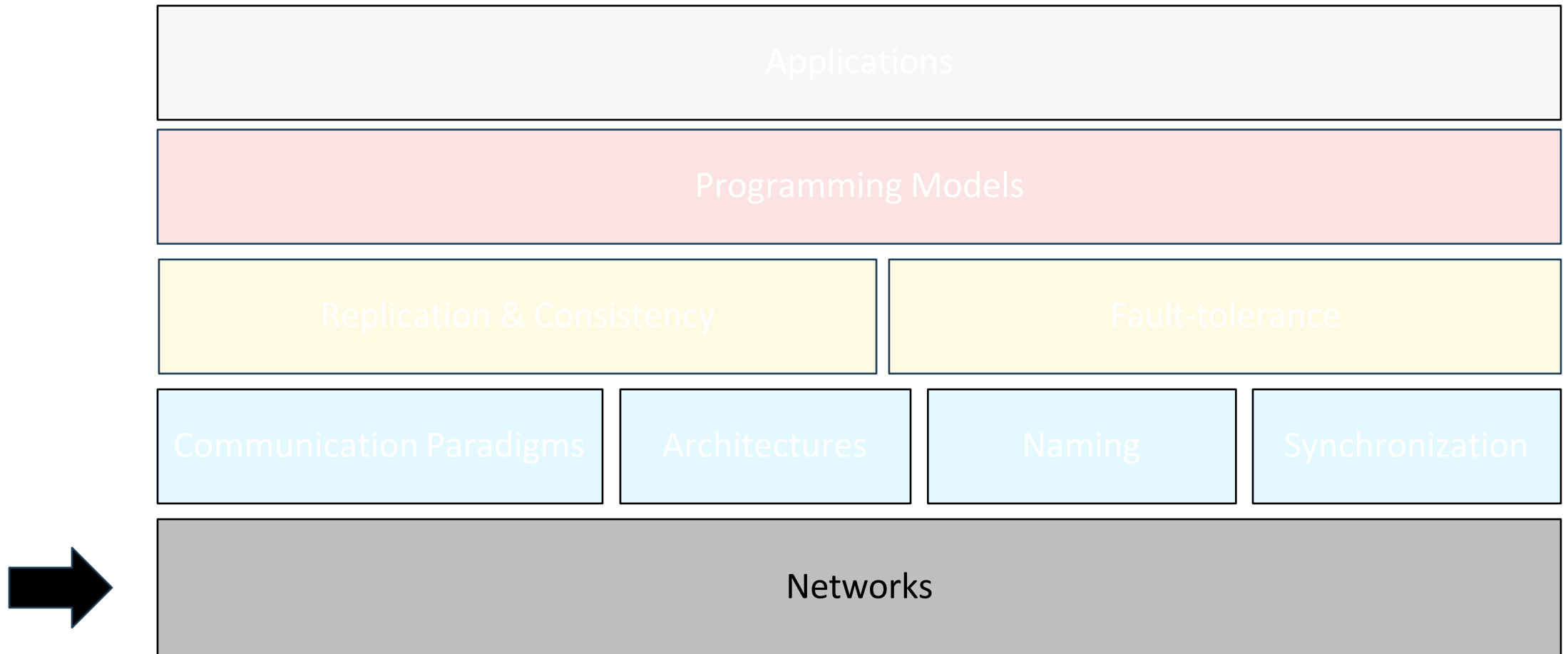


Having Such a System involves
addressing several key components
and challenges to ensure that the
system operates effectively and
efficiently

Course Map



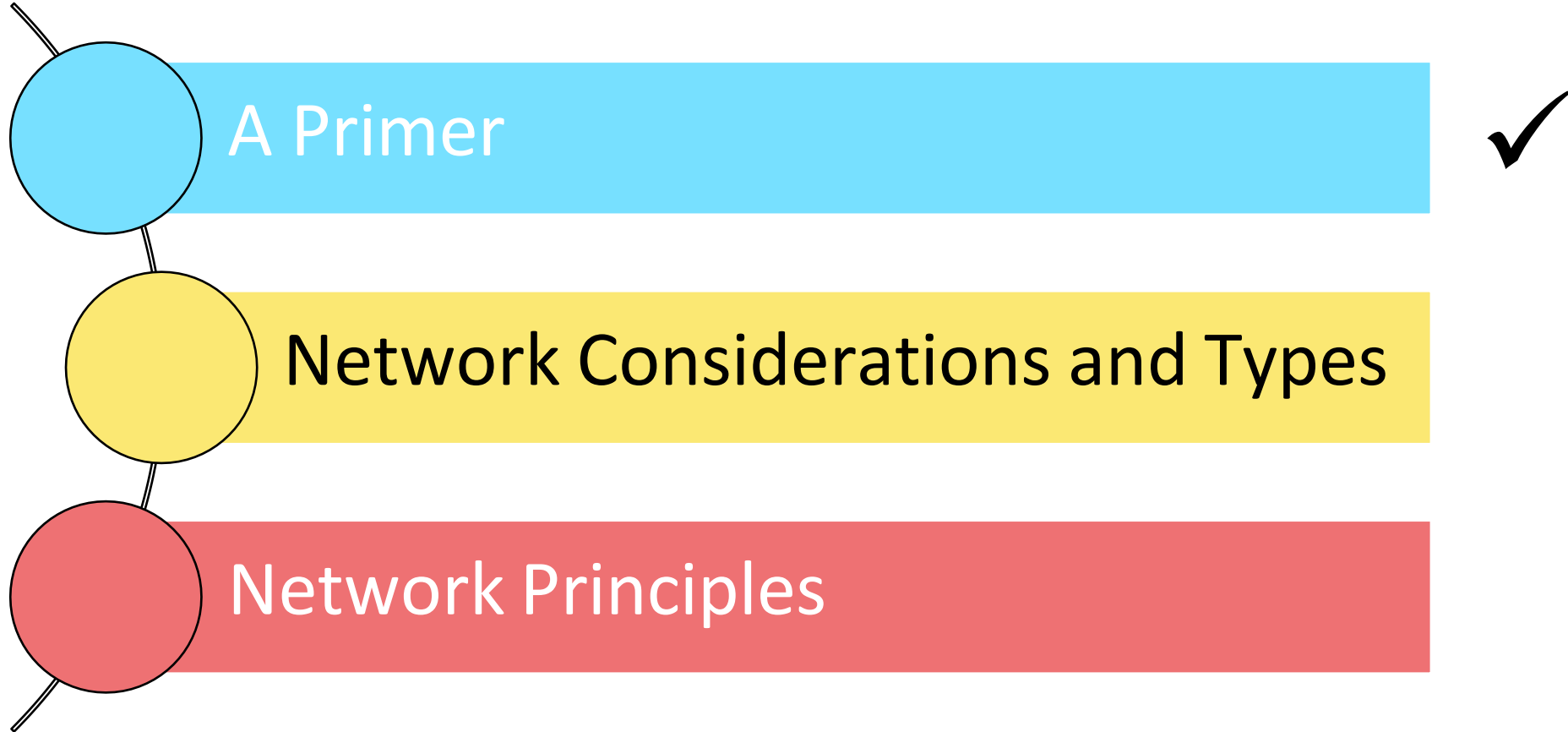
Course Map



Introduction to Networks – Learning Outcomes

- After two lectures on networks, you will be able to:
 - Identify different **types of networks**
 - Describe various **networking principles** such as layering, encapsulation, and packet-switching, among others
 - Examine how **packets** are **routed**
 - Realize how **congestion** is **controlled**
 - Analyze the **performance, scalability, and reliability** of networks

Outline



Networks in Distributed Systems

- A distributed system is a collection of components that **communicate** to solve a problem
- **Why should designers of distributed systems know about networks?**
 - Networking issues severely affect performance, fault-tolerance, and security of distributed systems
 - E.g., **Gmail** outage on Sep 1, 2010 – Google Spokesman said “*we had slightly underestimated the **load** which some recent changes placed on the request **routers**. few of the request routers became overloaded... causing a few more of them to also become overloaded, and **within minutes** nearly all of the request routers were overloaded.*”

Networks in Distributed Systems

- A distributed system is a collection of components that **communicate** to solve a problem
- **Why should designers of distributed systems know about networks?**
 - Networking issues severely affect performance, fault-tolerance, and security of distributed systems
 - E.g., **YouTube** outage on Oct 16, 2018: “There was a problem with the **network**”
 - E.g., **AWS** outage on Dec 7, 2021: “An impairment of **several network devices** in the US-EAST-1 Region caused **widespread errors** in all AWS services”

Distribution Gain

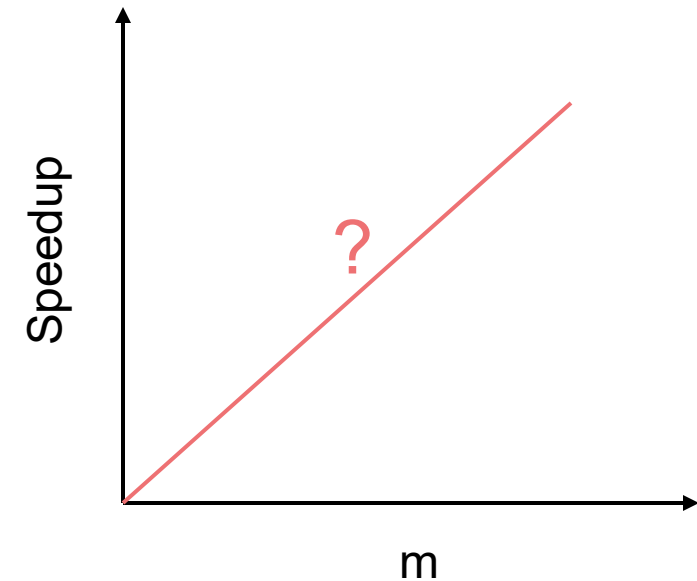
- To being with, why do we need distributed systems?
 - To run our programs faster (or more *efficient*)
 - Sometimes we have no option but to go distributed (Big Data applications)
- *How much faster will a distributed program run compared to its sequential counterpart?*

Distribution Gain

- Do we get a **linear** speedup as we distribute to more machines?

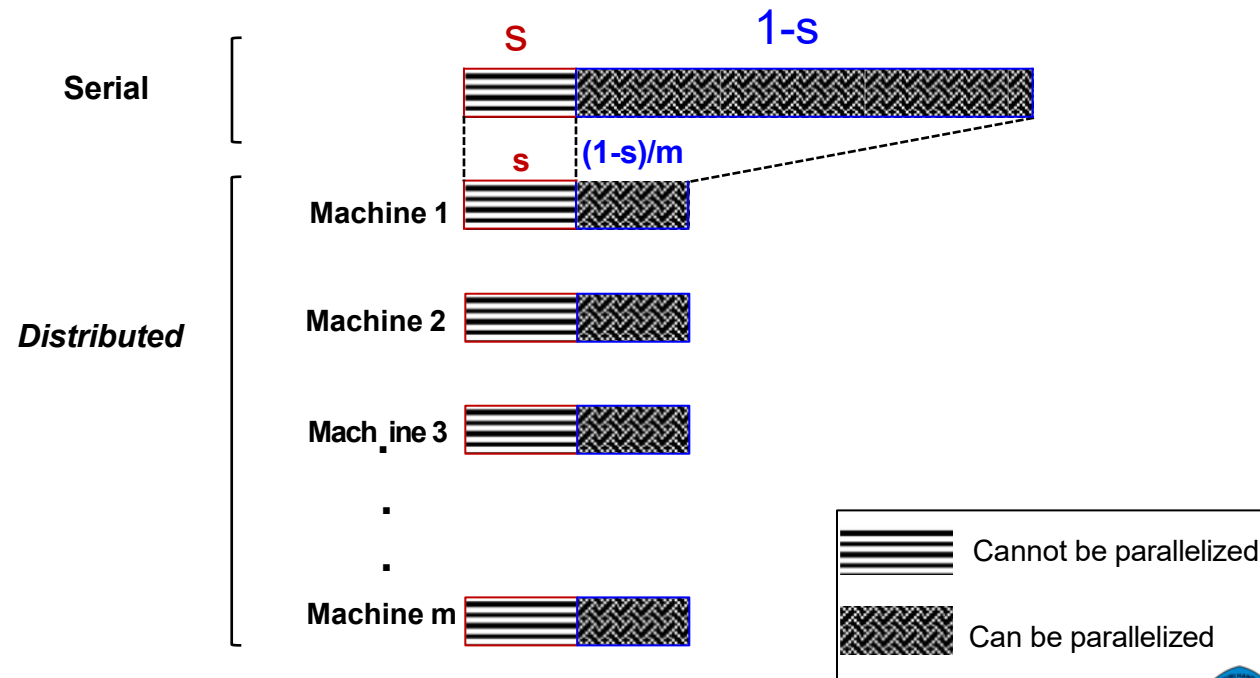
Suppose that the sequential execution of the program takes T_1 time units and the distributed execution on m machines takes T_m time units

$$Speedup = \frac{T_1}{T_m}$$



Amdahl's Law

Suppose that out of the entire execution of the program,
 s fraction of it is *not parallelizable* while $1-s$ fraction is *parallelizable*



Amdahl's Law

- Amdahl's Law formula:

$$Speedup = \frac{T_1}{T_m} = \frac{T_1}{(T_1 \times s + T_1 \times \frac{1-s}{m})} = \frac{1}{s + \frac{1-s}{m}}$$

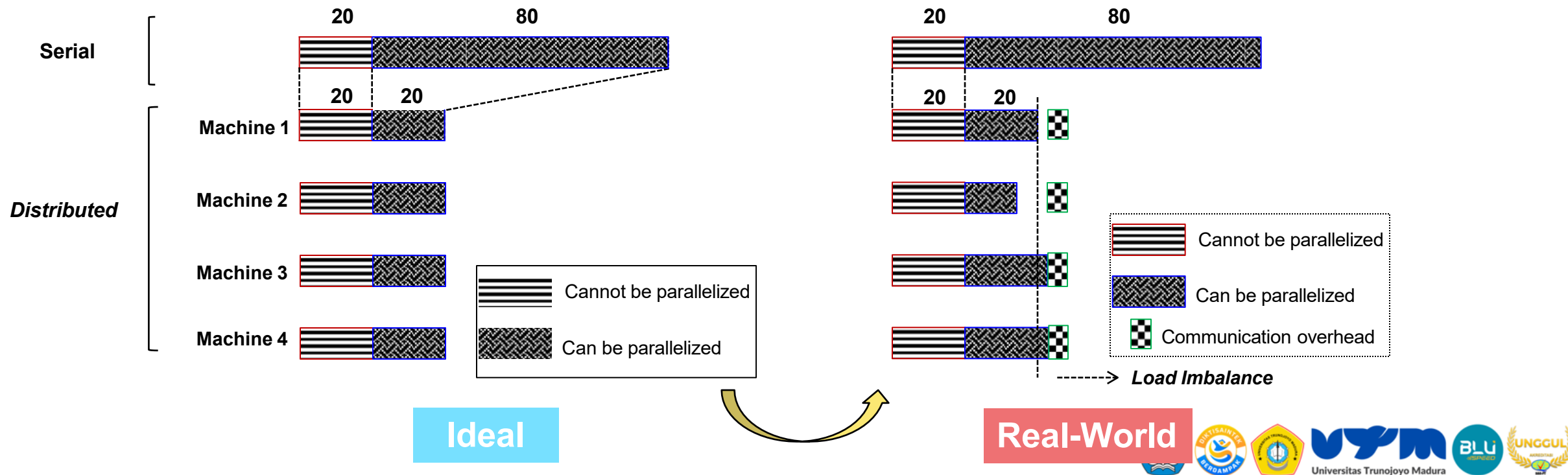
- Example:
 - Assume, 80% of your program is parallelizable
 - You run your program on 4 machines

$$Speedup = \frac{1}{s + \frac{1-s}{m}} = \frac{1}{0.2 + \frac{0.8}{4}} = 2.5$$

- Why did we get a *sub-linear* (i.e., 2.5x) and not a *linear* (i.e., 4x) speedup?

Amdahl's Law

- Amdahl's argument is too simplified
 - There is not only a part of the program that is not parallelizable, but also some **critical systems aspects** that are not captured in the law and may slow down the program



Guidelines

- How to improve the efficiency of your distributed program?
 - Maximize the fraction of the program that can be parallelized
 - Typically, inherent to the program
 - Maximize the efficiency of each machine
 - Scale up resources and/or optimize operations
 - **Fully utilize** (do not **underutilize** or **overutilize**) each machine
 - Choose the right number of machines (or **cluster size**)
 - Balance the workload across machines
 - **Heterogeneity** is not only caused by differences in the underlying platform, but also by resource availability (most critical on *shared* machines like on the Cloud)
 - Minimize the time spent on communication
 - **Latency** and **bandwidth**



Guidelines

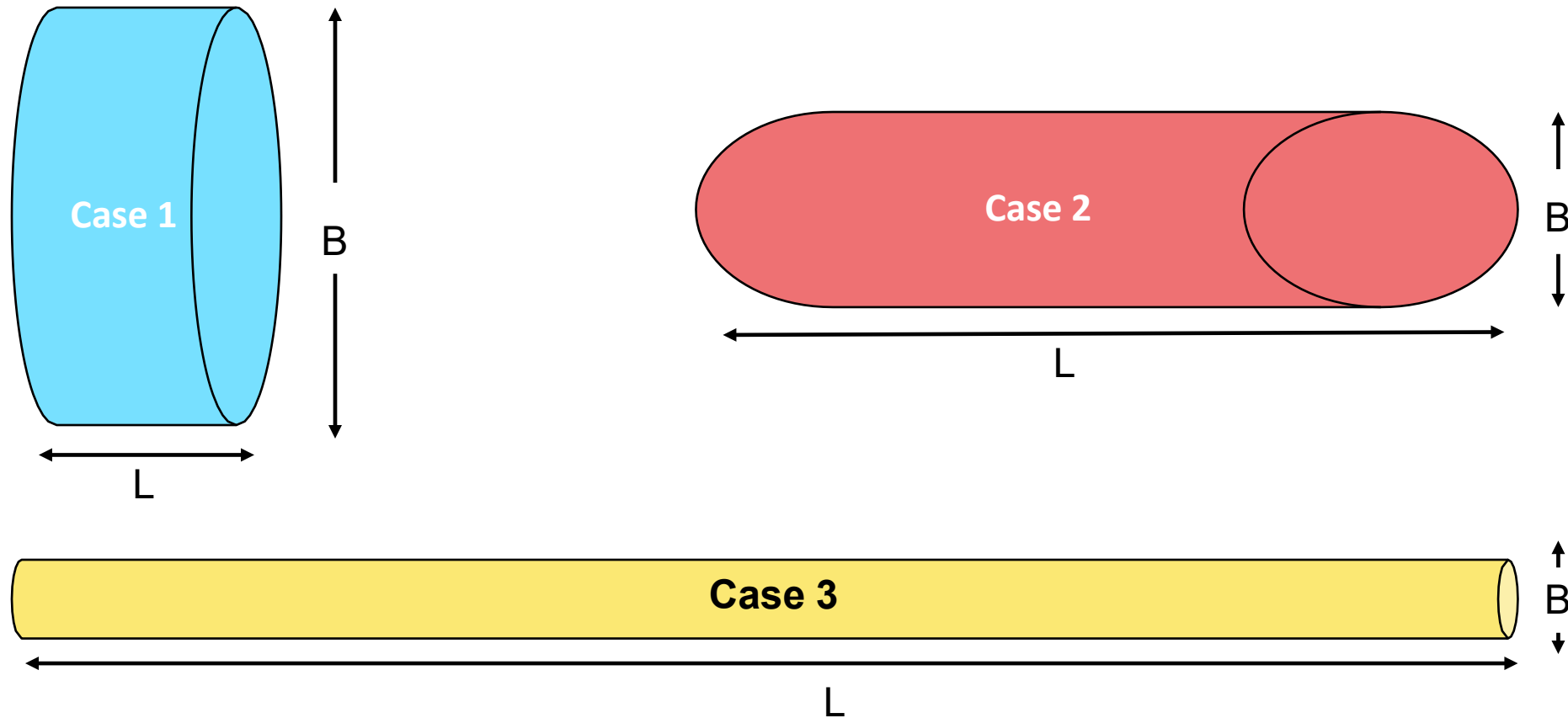
- In summary, be aware of:

- The **parallel** structures in your program
- The **efficiency** of each machine in your cluster
- The **utilization** of each machine in your cluster
- The **heterogeneity** across machines in your cluster
- The **latency** and **bandwidth** of your network

Amdahl's Law only considers this



Latency and Bandwidth



- B = Bandwidth (or *Capacity*) and L = Latency (or *Delay*)
- Bandwidth (often measured in bits per second or bytes per second)
- Latency (often measured in secs)

Message Transmission Time

$$\text{Message Transmission Time} = \text{Network Latency} + \frac{\text{Message Length}}{\text{Data Transfer Rate}}$$

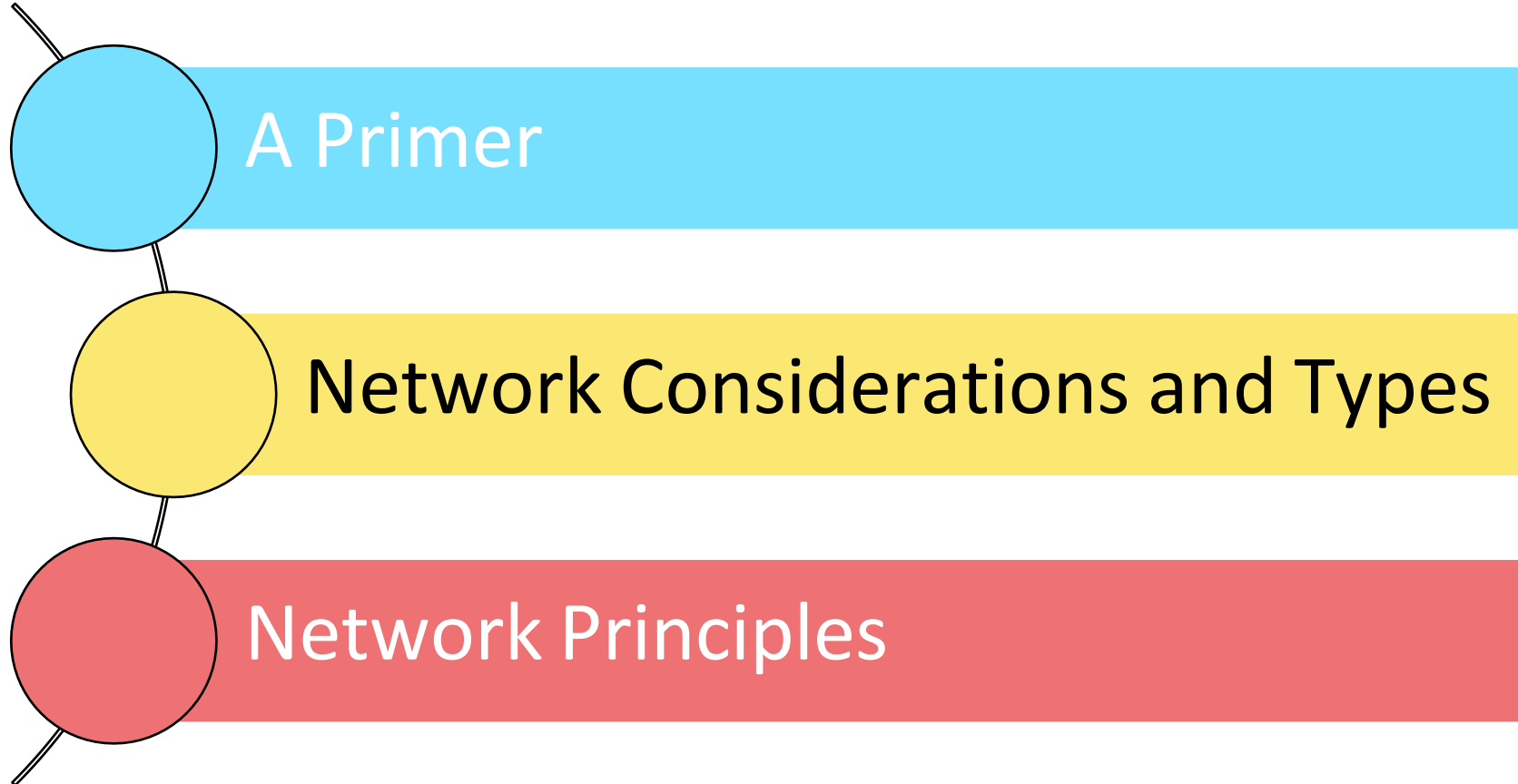
Delay introduced by the network itself, often due to routing, propagation delay, etc.

The **real** measure of the rate at which data is successfully transmitted over the network

VS

Bandwidth which sets the upper limit or the **potential maximum** rate at which data could be transmitted

Outline



Networks in Distributed Systems

Networking Issue	Comments on a Distributed System Design
Performance	Affects choices of whether to optimize for network or other resources
Scalability	Size of Internet is increasing; expect greater traffic and latency in future
Reliability	Detect communication errors and perform error-checks at the application layer (<i>end-to-end argument!</i>)
Security	Install firewalls at gateways; deploy end-to-end authentication; employ encryption, etc.,
Mobility	Expect intermittent connection for mobile devices
Quality-of-service	Internet is best-effort. It is hard to ensure strict QoS guarantees for, say, multimedia data

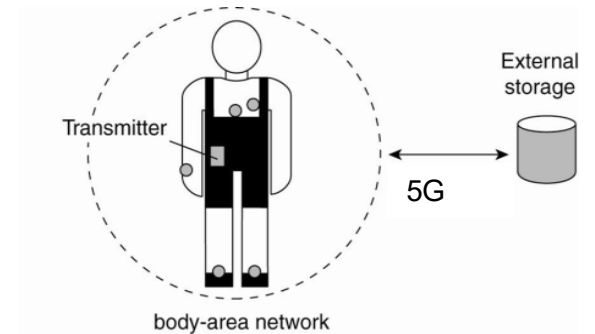


Network Classification

- Important ways to classify networks
 1. Based on size
 - Body Area Networks (BAN)
 - Personal Area Networks (PAN)
 - Local Area Networks (LAN)
 - Wide Area Networks (WAN)
 2. Based on technology
 - Ethernet Networks
 - Wireless Networks
 - Cellular Networks

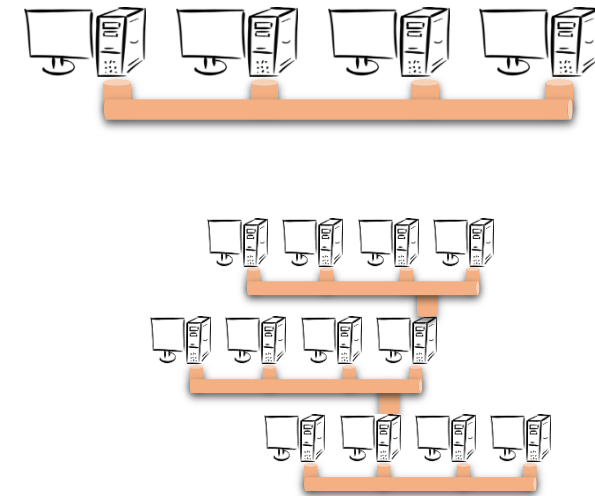
Network Classification – BANs and PANs

- Body Area Networks (BAN):
 - Devices form wearable computing units
 - Several Body Sensor Units (BSUs) communicate with Body Central Unit (BCU)
 - Typically, low-cost and low-energy networking
- Personal Area Networks (PAN):
 - PAN connects various digital devices carried by a user (mobile phones, tablets, cameras)
 - Low-cost and low-energy networking
 - E.g., Bluetooth



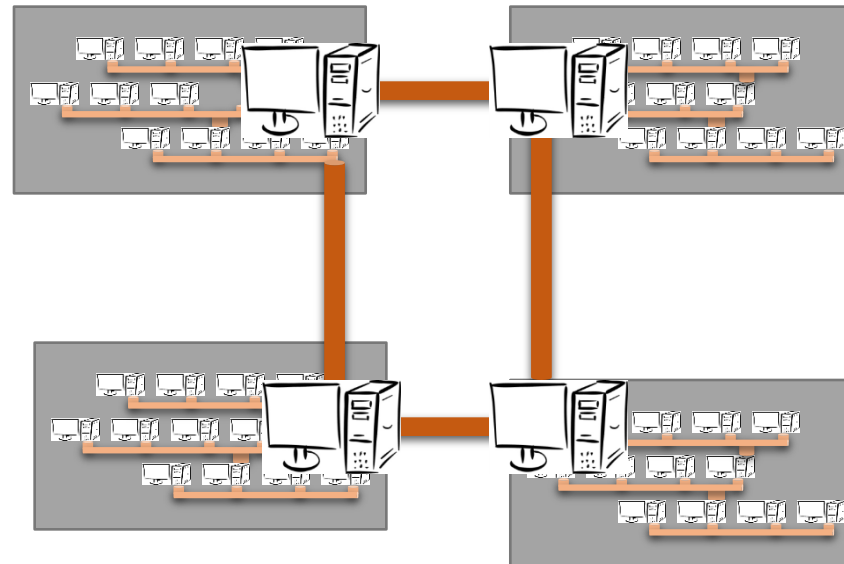
Network Classification – LAN

- Computers connected by single communication medium
 - E.g., Twisted copper wire, optical fiber
- High data-transfer-rate and low latency
- LAN consists of
 - Segment
 - Usually within a department/floor of a building
 - Shared bandwidth, no routing necessary
 - Local Networks
 - Serves campus/office building
 - Many segments connected by a switch/hub
 - Typically, represents a network within an organization

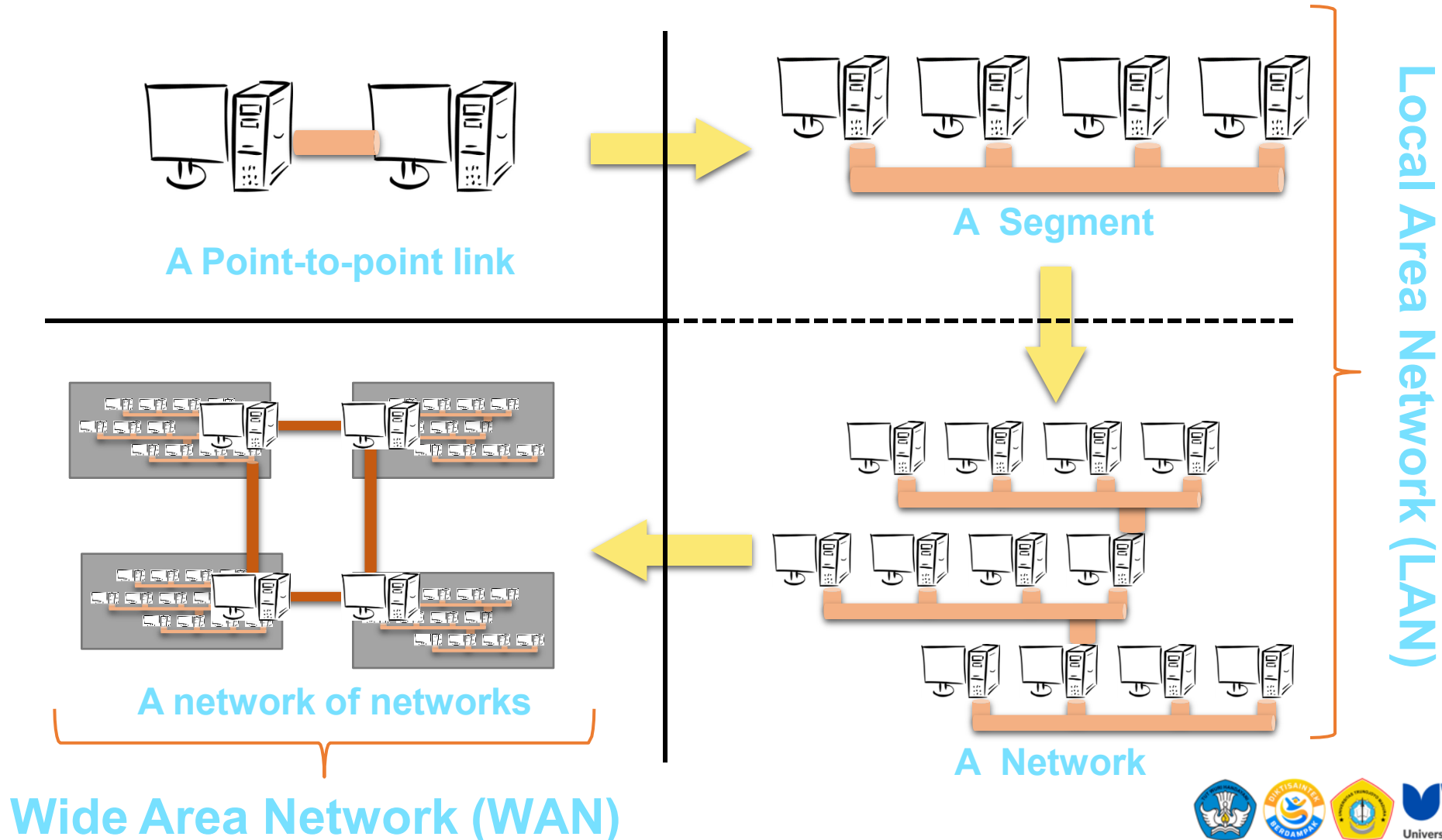


Network Classification – WAN

- Generally, covers a wider area (cities, countries,...)
- Consists of networks of different organizations
- Traffic is routed from one organization to another
 - Routers
- Bandwidth and latency
 - Vary
 - Worse than a LAN
- Largest WAN = Internet



Brief Summary of Important Networks (Based on Size)



Network Classification

- Important ways to classify networks

1. Based on size

- Body Area Networks (BAN)
- Personal Area Networks (PAN)
- Local Area Networks (LAN)
- Wide Area Networks (WAN)

2. Based on technology

- Ethernet Networks
- Wireless Networks
- Cellular Networks

Types of Networks – Based on Technology

- Ethernet Networks

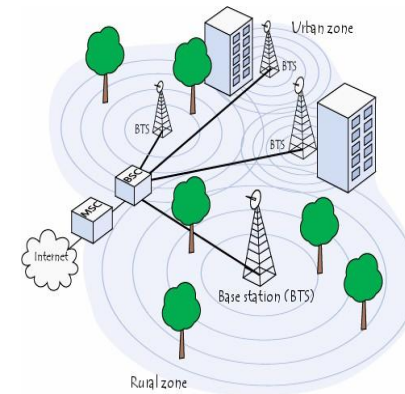
- Predominantly used in the wired Internet

- Wireless LANs

- Primarily designed to provide wireless access to the Internet
 - Low-range (100s of m), high-bandwidth

- Cellular networks (2G/3G/4G/5G)

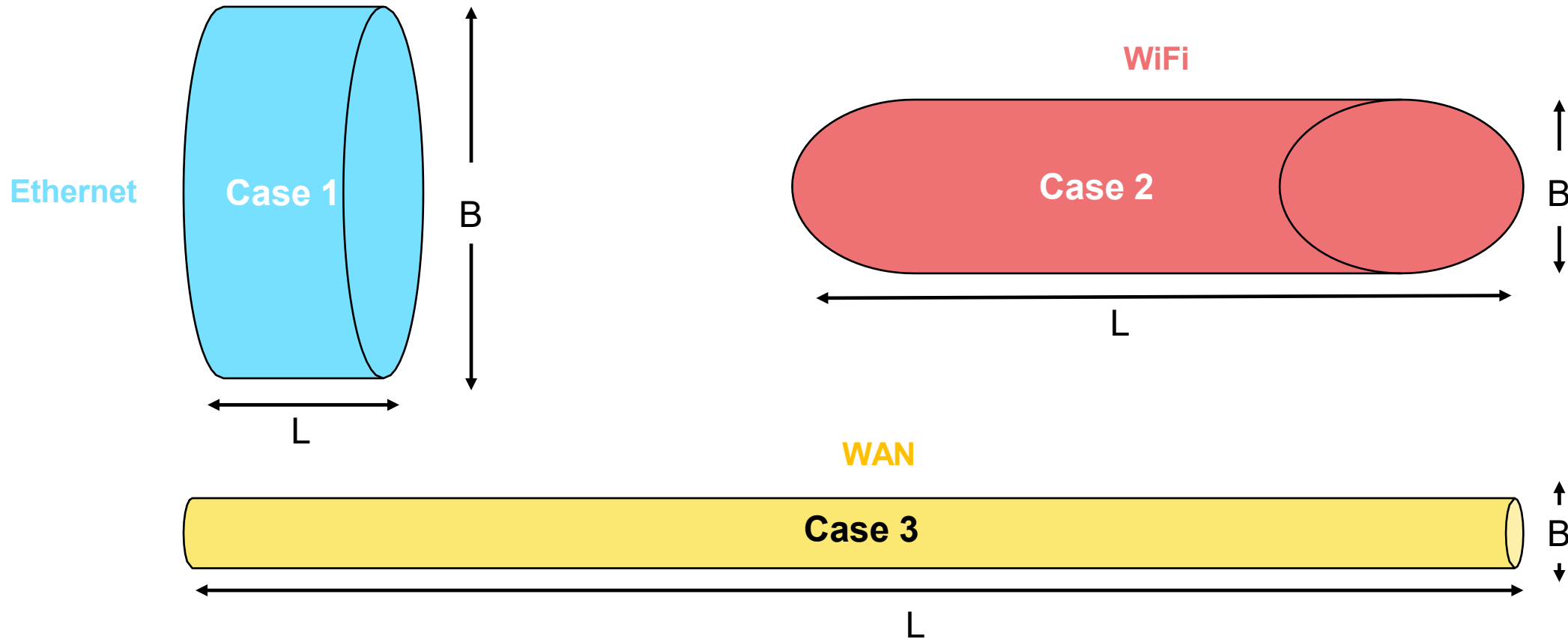
- Initially, designed to carry voice
 - Large range (few kms)
 - Low-bandwidth



Typical Performance for Different Types of Networks

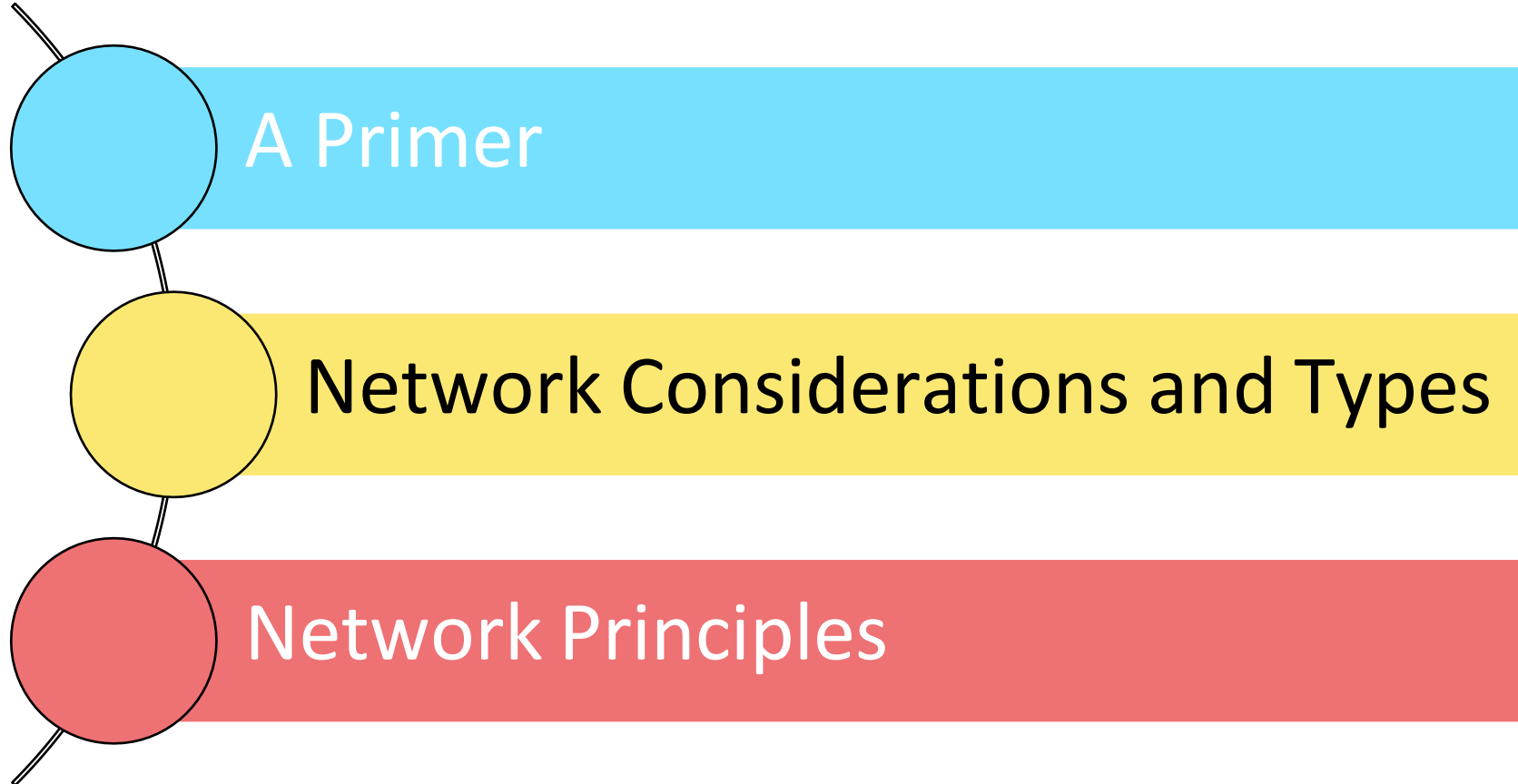
Network	Example	Range	Bandwidth (Mbps)	Latency (ms)
Wired LAN	Ethernet	1-2 km	10 – 10,000	1 – 10
Wired WAN	Internet	Worldwide	0.5 – 600	100 – 500
Wireless PAN	Bluetooth	10 – 30 m	0.5 – 2	5 – 20
Wireless LAN	WiFi	0.15 – 1.5 km	11 – 108	5 – 20
Cellular	2G – GSM	100m – 20 km	0.270 – 1.5	5
Cellular	3G	1 – 5 km	348 – 14.4	100 – 500
Cellular	4G	16 km	10- 100	36-48
Modern Cellular	5G	2 km	50- 1000	10-30

Latency and Bandwidth



- B = Bandwidth (or *Capacity*) and L = Latency (or *Delay*)
- $B \times L$ gives approximately the number of bits in flight
- As $B \times L$ increases, *uncertainty* increases (more bits might get lost)
- High value of $B \times L$ leads to “*Buffer Bloat*”

Outline



Networking Principles

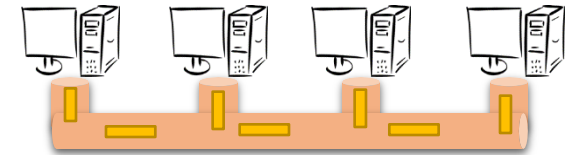
- Network Protocols
- Packet Transmission
- Network Layers
 - Physical layer
 - Data-link layer
 - Network layer and routing
 - Transport layer and congestion control

Networking Protocols

- If two entities want to communicate on a network, pre-defined agreements are necessary
 - How a message will be formatted?
 - How does the receiver know the last bit in the message?
 - How can a receiver detect if the message is damaged?
- “Protocol” is a well-known set of rules and formats to be used for communication between the entities
 - Implemented by software modules located in the sending and receiving computers
- Standardizing a well-known set of protocols supports communication among *heterogeneous* entities

Data Transmission

- Data (message) is broken into **packets** that are transmitted independently across a network (known as Packet Switching)
 - A packet is the unit of data that is transmitted between an origin and a destination
 - Packets can be of arbitrary lengths

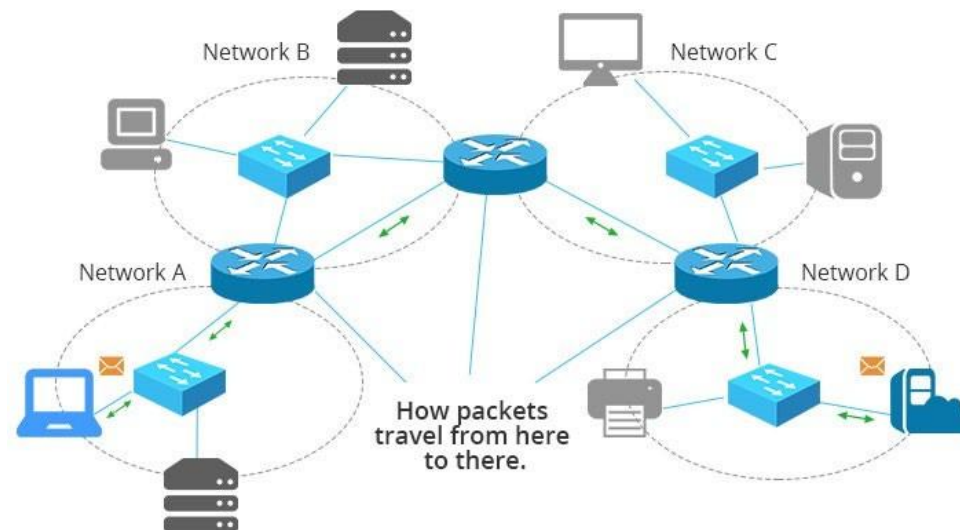


- Maximum size of the packet is known as Maximum Transmission Unit (MTU)
 - MTU prevents one host from sending a very long message
- Each packet has two main fields
 - Header: Contains meta-information about the packet
 - e.g., Length of the packet, receiver ID
 - Data



Networking Protocols - Functionalities

1. Defining how application data is formatted and processed
 - E.g. HTTP a web page request, SMTP for email
2. Dividing messages into packets before transmission, reassembling them at the receiver side and delivering messages to their relevant application process
3. Routing packets across WAN routers (hops) until they get to the sender's LAN
4. Transmitting packets in the LAN or the medium (e.g. Ethernet) to which the sender/receiver are attached
5. Turning bits into signals to be transferred over the communication medium (e.g. Ethernet)



Networking Protocols - Functionalities

*To ensure proper communication and functionality across the network, networking protocols **encapsulate** (pack) **data** with **headers** at the sender side and **decapsulate** (unpack) it at the receiver side*



Targeting a Packet to an Application Process

Analogy: Sending a package through mail to a person

The City and Postal Code “Springfield, IL 62704, USA”

Ensures that the *package (data)* is routed through various *cities and postal systems (networks)* to reach the *correct city (network)*.



Internet Protocol (IP) Address

(e.g. 142.250.190.78)

Street and Building Number “123 Maple Street, Building 4B”

Once in the *right neighborhood (local network)*, postal worker uses street and building number to deliver to the right *building (device)*



Media Access Control (MAC) Address

(e.g. 00:1A:2B:3C:4D:5E)

Person's Office Number “Office 1016”

Once delivered to the right *building (device)*, it needs to be delivered to the *right person (application process/service)*



Port Number

(e.g. 80 for HTTP, 22 for SSH)

Networking Protocols - Headers

To ensure proper communication and functionality across the network, networking protocols encapsulate **data** with headers at the sender side and decapsulate it at the receiver side

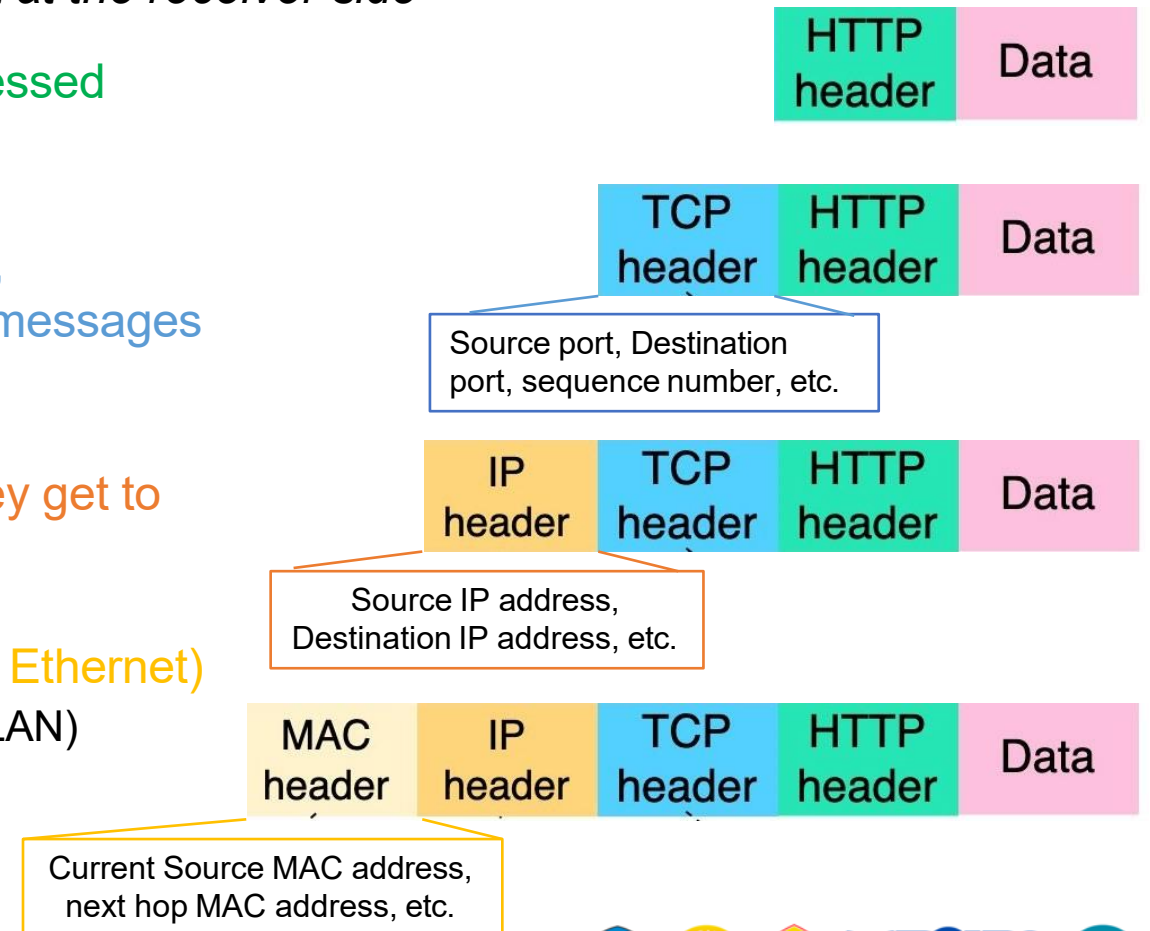
1. Defining how application data is formatted and processed
(E.g. HTTP for a web page request, SMTP for email)

2. Dividing messages into packets before transmission, reassembling them at the receiver side and delivering messages to their relevant application process (E.g. TCP, UDP)

3. Routing packets across WAN routers (hops) until they get to the sender's LAN (E.g. IP, ICMP)

4. Transmitting packets in the LAN or the medium (e.g. Ethernet) to which the sender/receiver are attached (E.g. ARP, VLAN)

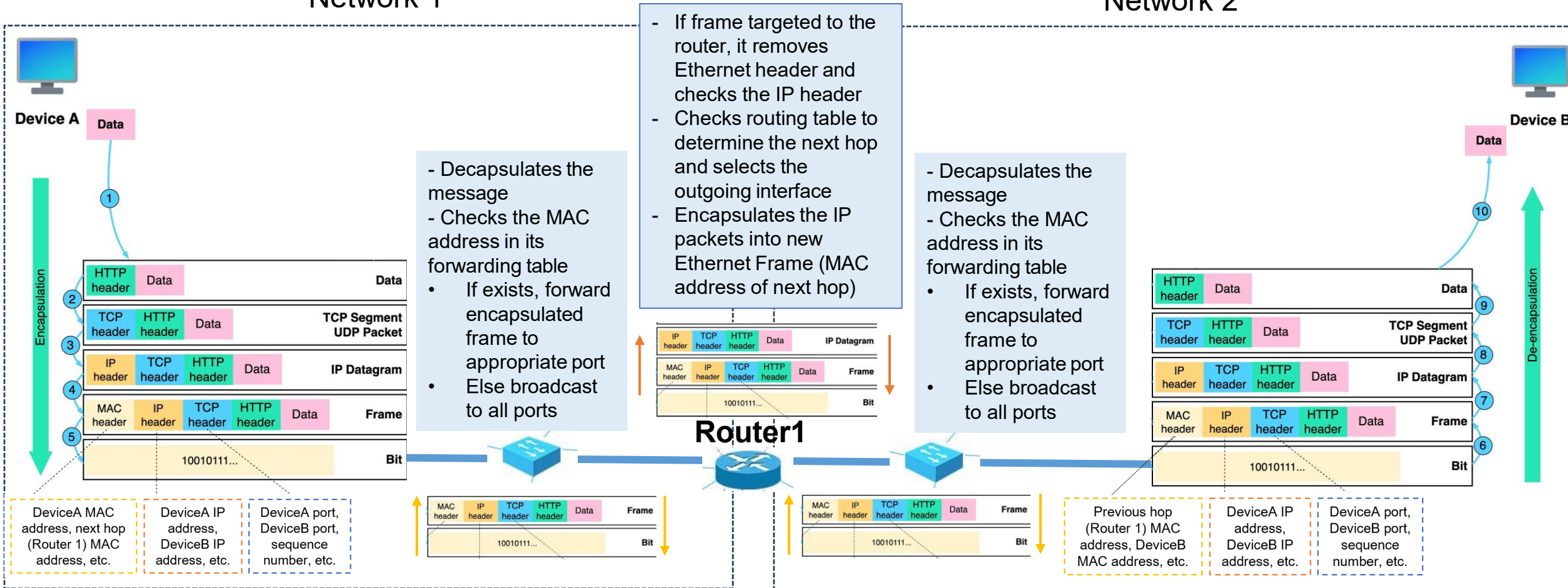
5. Turning bits into signals to be transferred over the communication medium (e.g. Ethernet)



Packet Lifetime

Network 1

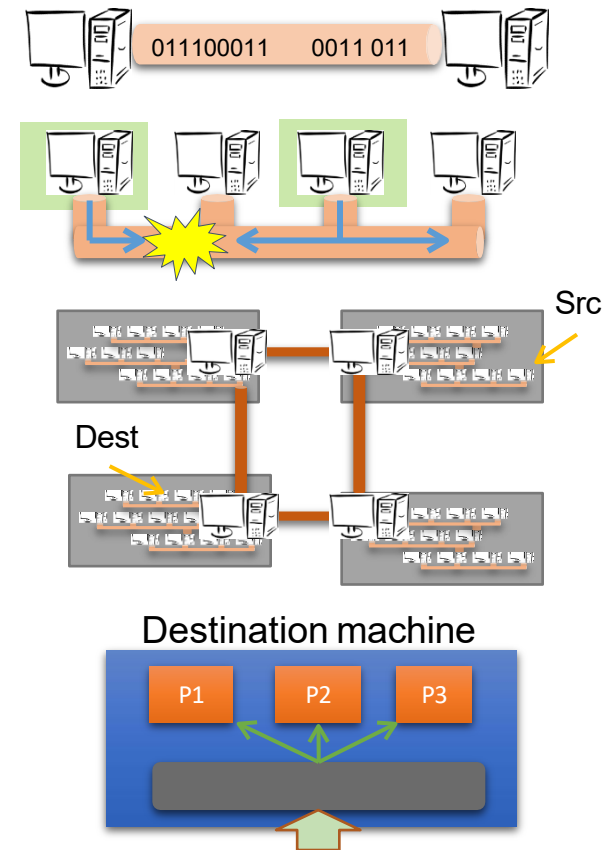
Network 2



Network Layers

- Network software is arranged into a hierarchy of layers
 - Protocols in one layer perform one specific functionality
 - Layering is a scalable & modular design for complex software
- Typical functionalities in a network software:

Functionality	Layer
Transmits bits over a transmission medium	Physical
Coordinates transmissions from multiple hosts that are directly connected over a common medium	Data link
Routes the packets through intermediate networks	Network
Handles messages – rather than packets – between sender and receiver processes	Transport
Satisfies communication requirements for specific applications	Application

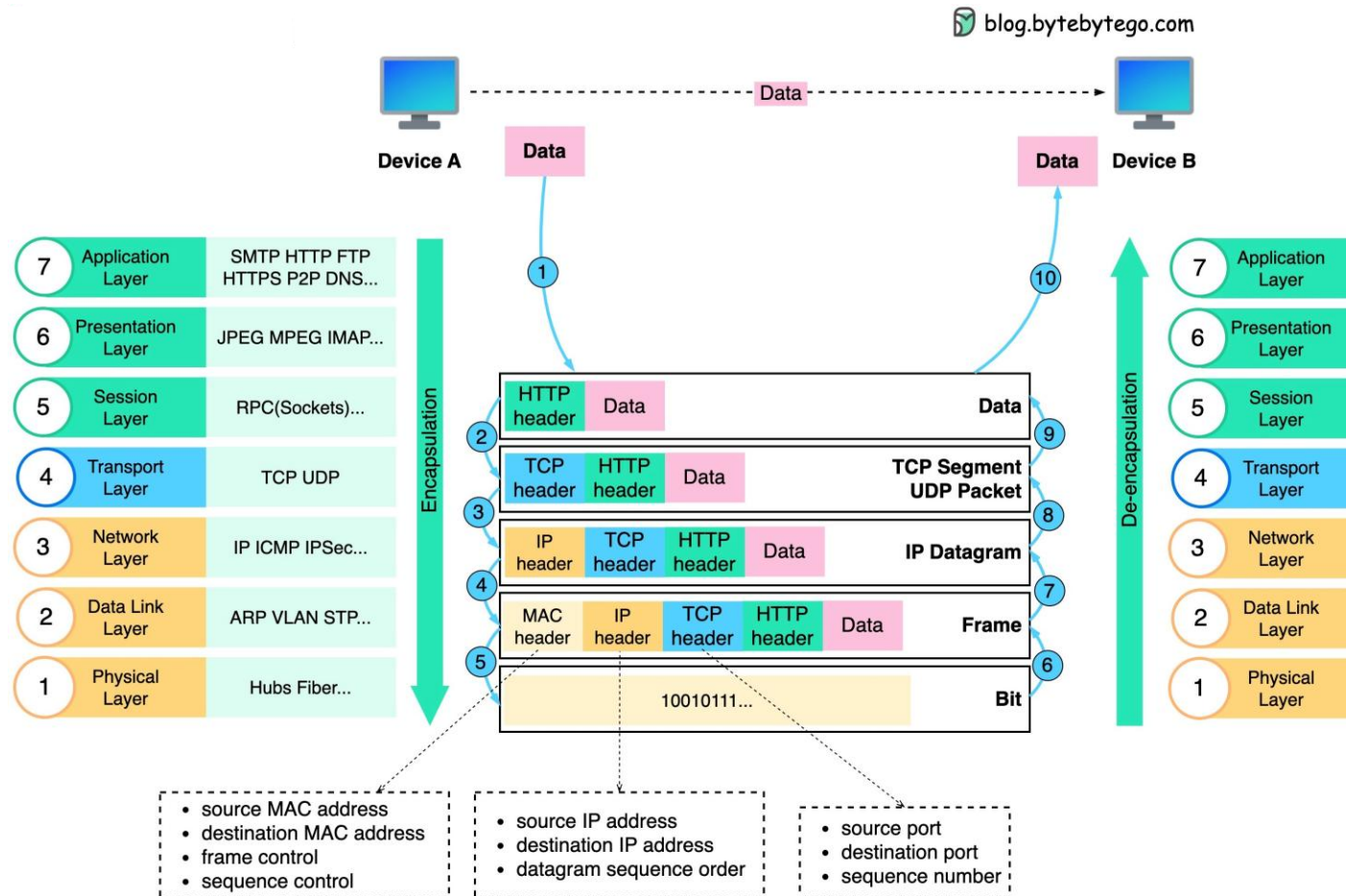


OSI Reference Model

- Open Systems Interconnection (OSI) Reference Model
 - A layered networking model standardized by ISO
 - The model identifies various layers and their functionalities

Functionality	Layer	Example Protocols
Satisfy communication requirements for specific applications	Application	HTTP, FTP
Transmit data in network representation that is independent of representation in individual computers	Presentation	CORBA data representation
Support reliability and adaptation, such as failure detection and automatic recovery	Session	SIP
Handle messages – rather than packets – between sender and receiver processes	Transport	TCP, UDP
Route the packet through intermediate networks	Network	IP, ATM
Coordinate transmissions from multiple hosts that are directly connected over a common medium	Data-link	Ethernet MAC
Transmit bits over a transmission medium	Physical	Ethernet

OSI Reference Model



Layers that We Will Study Today

1. Physical layer
2. Data-link layer
3. Network layer

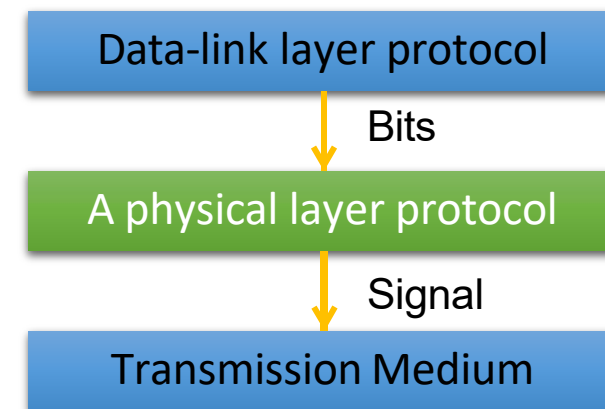
Layers that We Will Study Today

1. Physical layer
2. Data-link layer
3. Network layer

Physical Layer

- Physical layer protocols transmit a sequence of bits over a transmission medium
 - Modulate the bits into signals that can be transmitted over the medium

Transmission Medium	Type of signal transmitted
Twisted-pair (Ethernet cable)	Electrical signal
Fiber Optic Circuits	Light signal
Wireless channel	Electro-magnetic signal



Layers that We Will Study Today

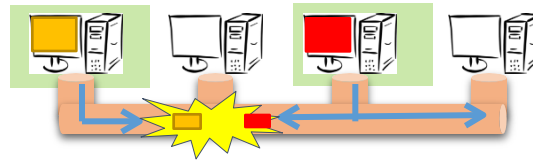
1. Physical layer
2. Data-link layer
3. Network layer

Data-link Layer

- Protocols in data-link layer ensure that the packets are delivered from one host to another within a local network
- Data-link layer protocols provide two main functionalities:
 - How to coordinate between the transmitters such that packets are successfully received?
 - Coordination
 - How to identify another host on the local network?
 - Addressing over local networks

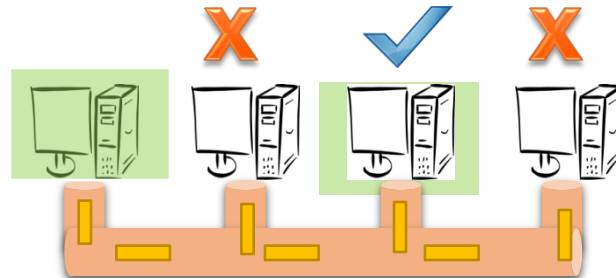
Coordination at Data-link Layer

- A packet is not received successfully at the receiver if a sender transmits the data when another sender's transmission is active
 - The packet is said to have experienced **collision** if it is not successfully received at the receiver
- Collision is avoided by sensing the medium before transmission



Addressing over Local Networks

- Each device that is connected to a network has a unique address called **Medium Access Control (MAC) address**
 - MAC addresses are six bytes long
 - e.g., 2A:D4:AB:FD:EF:8D
- Approach:
 - Data-link layer *broadcasts* the packet over the medium
 - (FF-FF-FF-FF-FF-FF is used for broadcasting).
 - Receiver reads the packet header and checks if the packet is addressed to it



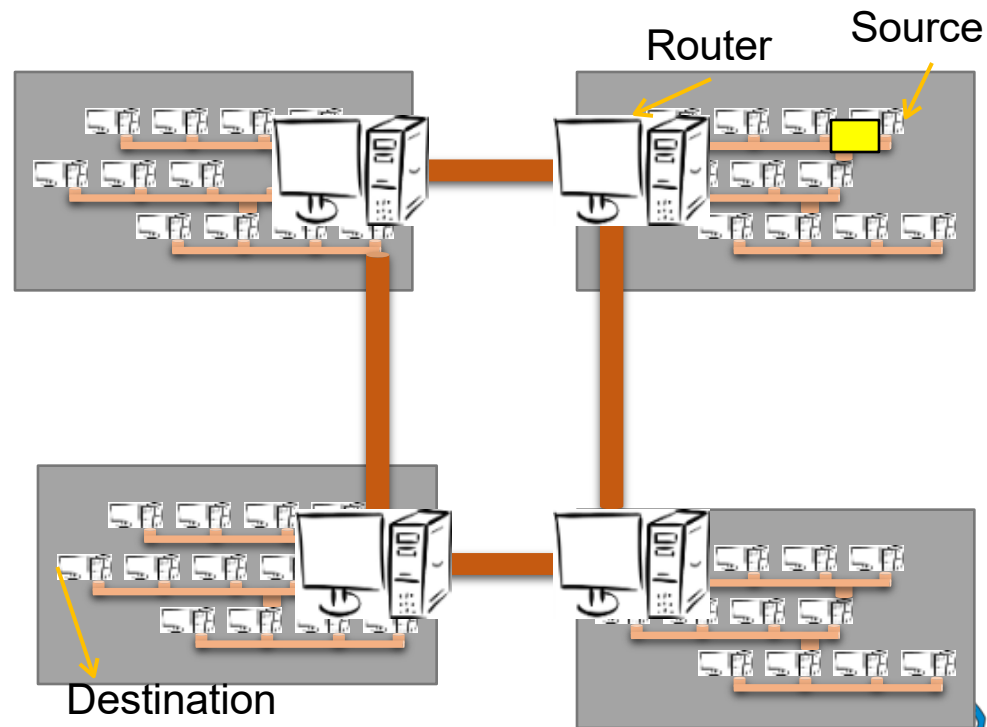
Layers that We Will Study Today

1. Physical layer
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Network Layer

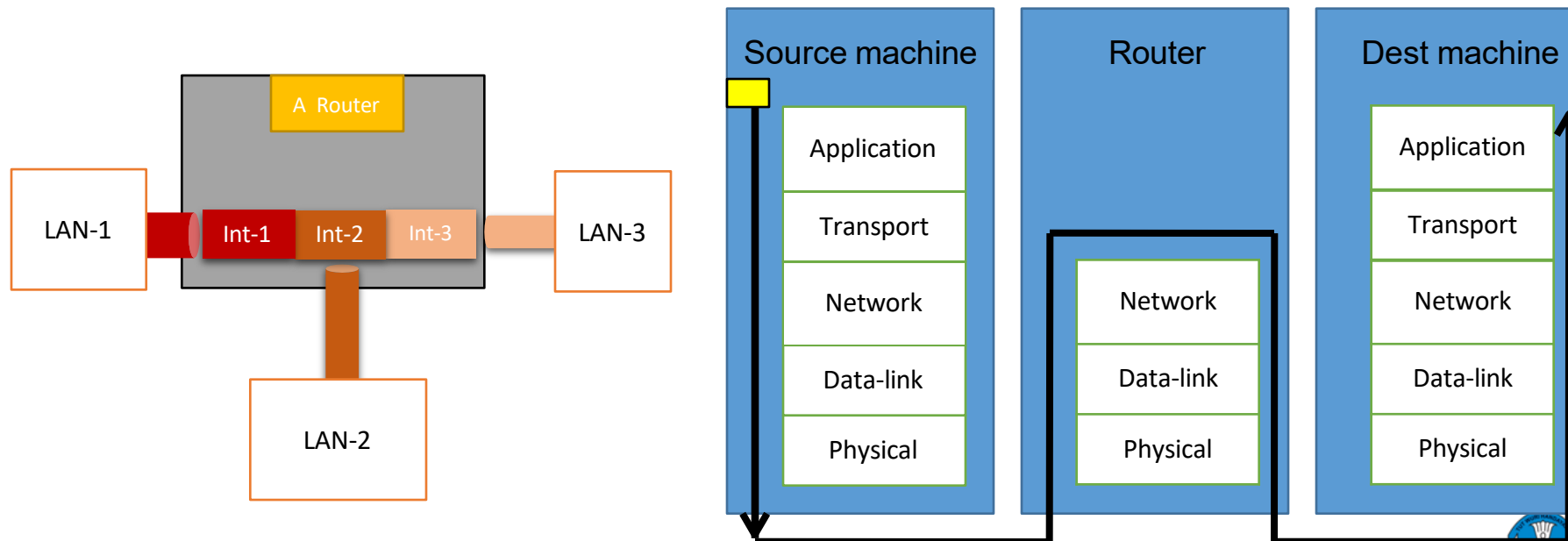
- Network layer protocols perform the role of **routing**
 - They ensure that a packet is routed from the source machine to the destination machine
 - Packets may traverse different LANs to reach the destination

- Internet Protocol (IP) is a widely-used network layer protocol
 - IP addresses are typically used to identify machines



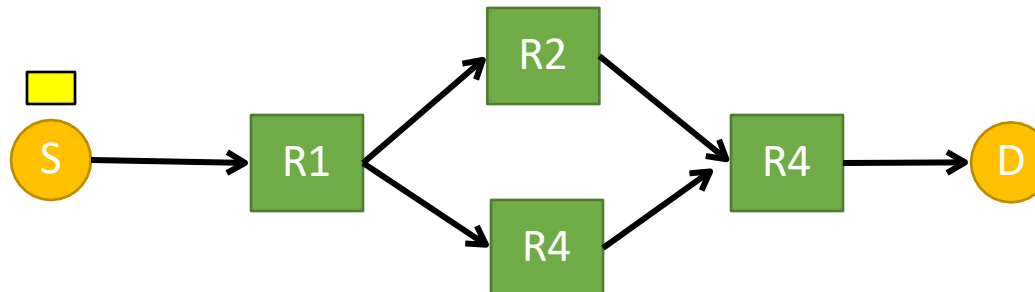
Router

- A **router** is a device that forwards the packets between multiple networks
- Routers are connected to two or more networks
 - Each network *interface* is connected to a LAN or a host
- Packet travels up until the network layer on the router



Routing Algorithm

- Packets have to be transmitted in a series of hops through the routers
 - The *series of hops* that a packet takes is known as a **route**
- **Routing algorithm** is responsible for determining the routes for the transmission of packets
- Challenges for designing routing algorithms in the Internet:
 - Performance: The *traffic* across different networks vary
 - Router *failures*: Routers in the Internet may fail



Routing Algorithm (Cont'd)

- Routing algorithms have two activities
 1. Determine the *next-hop* taken by each packet
 - The algorithm should be fast and efficient
 2. Dynamically update *connectivity information*
 - Maintain the knowledge of the network by monitoring routers and traffic
- The above activities are *distributed* throughout the network
 - Routing decisions are made on an *hop-by-hop basis*
 - Information about possible next-hop routers is stored locally
 - Information is updated periodically
- We will study in the next lecture a simple routing algorithm called “Distance Vector Algorithm”

Distance Vector Algorithm

Distance Vector (DV) relies on graph theory to find the best route in a given network

It uses a well-known shortest path algorithm called ***Bellman-Ford***

Two activities for the DV routing algorithm:

1. Determining the best next-hop at each router
2. Dynamically update connectivity information at all the routers

Distance Vector Algorithm –

1) Next-hop Determination

Each router maintains a **routing table** that consists of:

Destination: The destination IP of the packet

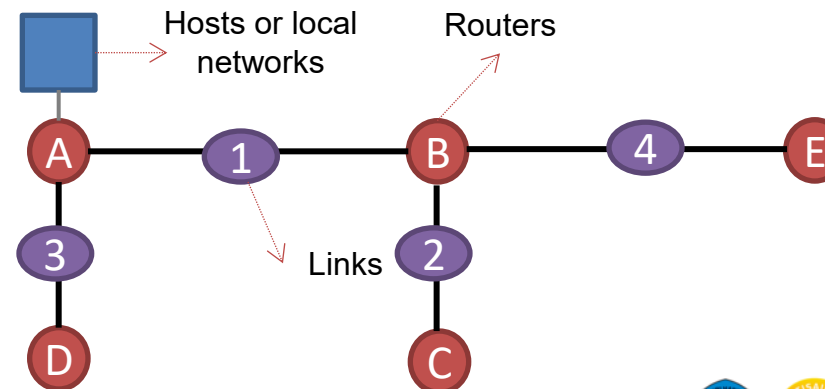
Link: The outgoing link (or interface) on which the packet should be forwarded

Cost: The *distance* between the router and the destination

E.g., Cost can be estimated as the delay for the packet to reach the destination

Router looks up the table to determine the best next-hop

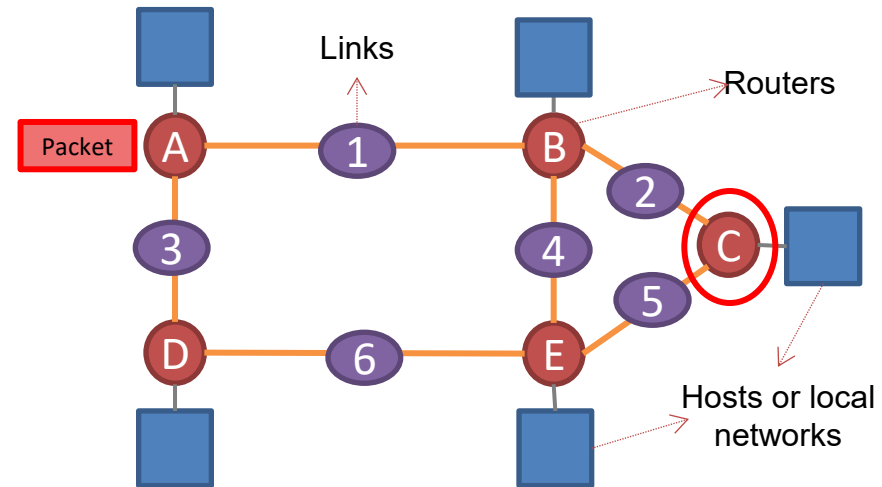
<i>Routing table at a router A</i>		
<i>To</i>	<i>Link</i>	<i>Cost</i>
A	local	0
B	1	1
C	1	2
D	3	1
E	1	2



Routing Tables Lookup-Example Scenario

Routings from A		
To	Link	Cost
A	local	0
B	1	1
C	1	2
D	3	1
E	1	2

Routings from B		
To	Link	Cost
A	1	1
B	local	0
C	2	1
D	1	2
E	4	1



Routings from C		
To	Link	Cost
A	2	2
B	2	1
C	local	0
D	5	2
E	5	1

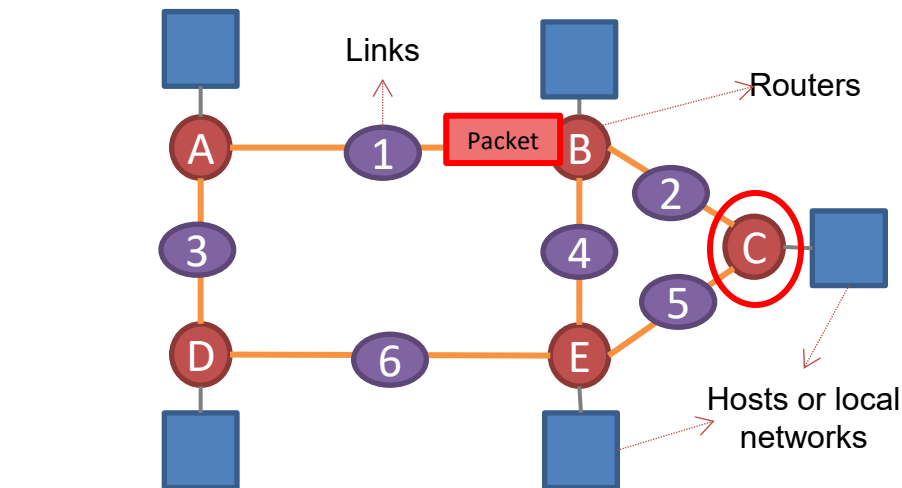
Routings from D		
To	Link	Cost
A	3	1
B	3	2
C	6	2
D	local	0
E	6	1

Routings from E		
To	Link	Cost
A	4	2
B	4	1
C	5	1
D	6	1
E	local	0

Routing Tables Lookup-Example Scenario

Routings from A		
To	Link	Cost
A	local	0
B	1	1
C	1	2
D	3	1
E	1	2

Routings from B		
To	Link	Cost
A	1	1
B	local	0
C	2	1
D	1	2
E	4	1



Routings from C		
To	Link	Cost
A	2	2
B	2	1
C	local	0
D	5	2
E	5	1

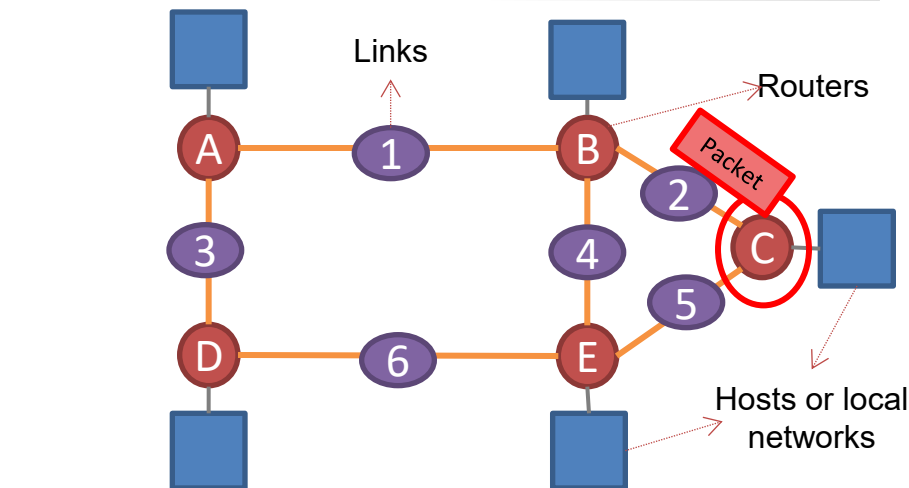
Routings from D		
To	Link	Cost
A	3	1
B	3	2
C	6	2
D	local	0
E	6	1

Routings from E		
To	Link	Cost
A	4	2
B	4	1
C	5	1
D	6	1
E	local	0

Routing Tables Lookup-Example Scenario

Routings from A		
To	Link	Cost
A	local	0
B	1	1
C	1	2
D	3	1
E	1	2

Routings from B		
To	Link	Cost
A	1	1
B	local	0
C	2	1
D	1	2
E	4	1



Routings from C		
To	Link	Cost
A	2	2
B	2	1
C	local	0
D	5	2
E	5	1

Routings from D		
To	Link	Cost
A	3	1
B	3	2
C	6	2
D	local	0
E	6	1

Routings from E		
To	Link	Cost
A	4	2
B	4	1
C	5	1
D	6	1
E	local	0

Distance Vector Algorithm – 2) Updating the Connectivity Information

Connectivity is updated by exchanging routing tables

Router Information Protocol (RIP) is used for sending update messages

1. Send a routing table to neighboring routers
Periodically, or when local table changes
2. When a neighbor's routing table is received:

Case	If the received routing table ...	Updates to the local routing table
1	Has a new destination that is not in the local routing table	Update the Cost and Link
2	Has a better-cost route to a destination in the local routing table	Update the Cost
3	Has a more recent information	Update the Cost and Link

Pseudocode for RIP

Send: Each t seconds or when TI changes, send TI on each non-faulty outgoing link

Receive: Whenever a routing table Tr is received on link n :

```

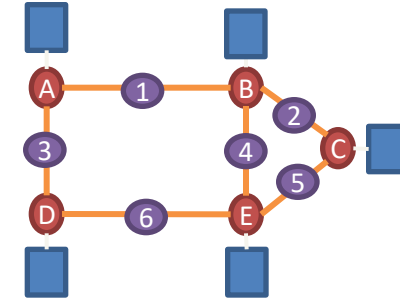
for all rows  $Rr$  in  $Tr$  {
  if ( $Rr.link \neq n$ ) {
     $Rr.cost = Rr.cost + 1$ ; // Update cost
     $Rr.link = n$ ; // Update next-hop
    if ( $Rr.destination$  is not in  $TI$ ) {
      add  $Rr$  to  $TI$ ; // add new destination to  $TI$ 
    }
  }
  else for all rows  $RI$  in  $TI$  {
    if ( $Rr.destination = RI.destination$ ) {
      //  $Rr.cost < RI.cost$ : remote node has better route
      //  $RI.link = n$ : information is more recent
      if ( $Rr.cost < RI.cost$  OR  $RI.link = n$ ) {
         $RI = Rr$ ;
      }
    }
  }
}

```

Case 1

Case 2

Case 3



TI at A

Routing table at router A		
To	Link	Cost
A	local	0
D	3	1
C	3	3

Tr recvd @ A from B on link $n=1$

Routing table of router B		
To	Link	Cost
A	1	1
B	local	0
C	2	1

Summary: Routing over Internet

Each machine over the Internet is identified by an IP Address

Source machine transmits the packet over its local network

Intermediate routers examine the packet, and forward it to the best next-hop router

If the destination is directly attached to the local network of a router, the router forwards the packet over the respective local network

Routers exchange information to keep an up-to-date information about the network

The Four Layers We Are Studying

1. Physical layer
2. Data-link layer
3. Network layer
4. Transport layer

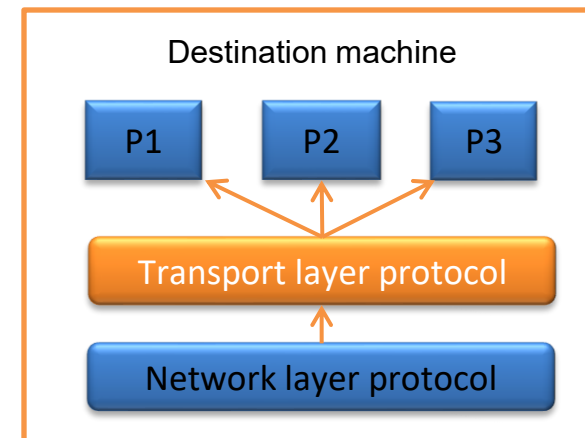
Transport Layer

Transport layer protocols provide end-to-end communication for applications

This is the lowest layer where messages (rather than packets) are handled

Messages are addressed to communication **ports** attached to the processes

Transport layer **multiplexes** each
packet received to its respective port



Simple Transport Layer Protocols

Simple transport protocols provide the following services:

1. Multiplexing Service
2. Connection-less Communication: The sender and receiver processes do not initiate a connection before sending the message
Each message is encapsulated in a packet (also called as *datagram*)
Messages at the receiver can be in different order than the one sent by the sender
E.g., User Datagram Protocol (*UDP*)

Advanced Transport Layer Protocols

Advanced transport layer protocols typically provide more services than simple multiplexing

Transmission Control Protocol (TCP) is a widely-used protocol that provides three additional services:

1. Connection-oriented Communication
2. Reliability
3. Congestion Control

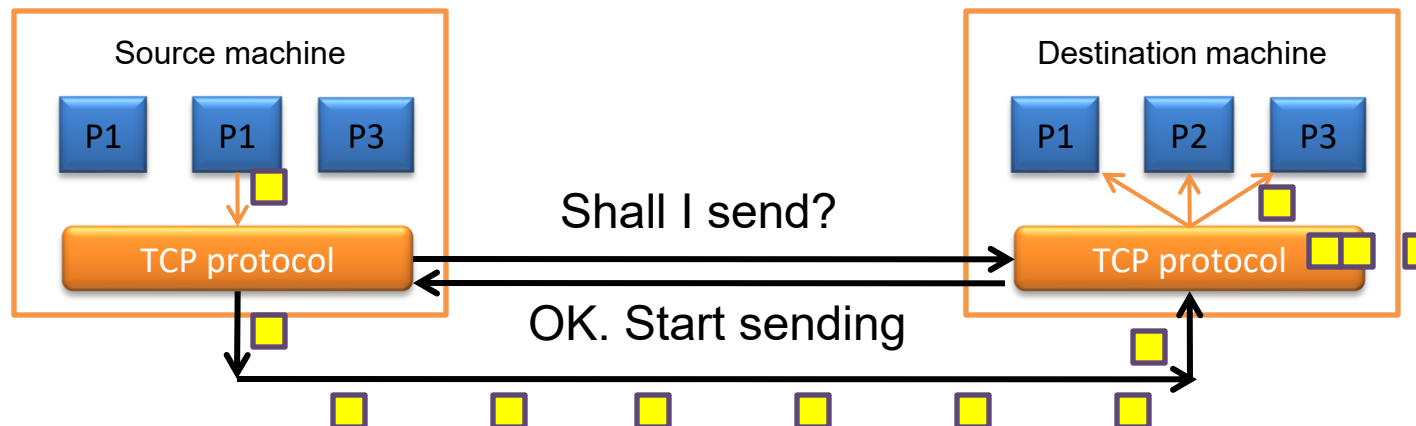
1. Connection-Oriented Communication

Sender and receiver will handshake before sending the messages

- Handshake helps to set-up connection parameters, and to allocate resources at destination to receive packets

Destination provides *in-order delivery* of messages to the intended process

- Destination will buffer the packets until previous packets are received
- It will then deliver packets to the process in the order that the sender had used



2. Reliability

Packets may be lost in the network due to buffer overflows at the router or transmission error(s)

In TCP, destination sends an ACK to the sender

- If ACK is not received at the sender, the sender will infer a packet error, and retransmit the packet

3. Congestion Control

The capacity of a network is limited by the individual communication links and routers

- Limited buffer space and link-bandwidth

What happens if a source transmits packets at a rate that is greater than the capacity of the network?

- Packets drop at intermediate routers
- Corresponding ACKs will NOT be received at the source
- The source retransmits
- More packets build-up on the router queue
- The network collapses

3. Congestion Control (Cont'd)

To avoid congestion, *two* functionalities can be adopted

1. Detect congestion at routers

If a router expects a buffer overflow, it typically follows one of two strategies:

- It drops packets and lets sources regulate upon observing packet losses
- It sends an “Explicit Congestion Notification (ECN)” packet to sources

2. Regulate input at sources

If the TCP-sender concludes congestion (e.g., it receives an ECN packet), then it reduces its sending rate

Recap: Introduction to Networking – Learning Objectives

You will identify how computers over the Internet communicate

Specifically, after the lecture in networking you will be able to:

- Identify different types of networks
- Describe networking principles such as layering, encapsulation, and packet-switching
- Examine how packets are routed and how congestion is controlled
- Analyze scalability, reliability, and fault-tolerance over the Internet

Next Class...

Remote Procedure Calls- Part I